COI-AT3

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1. Intelligent agents in a smart home automation system?

1. Introduction

Artificial intelligence [AI] is a branch of computer science that enables machines to learn, reason, solve problems, and make decisions. An intelligent agent is an important concept in AI.

An intelligent agent is an entity that perceives its environment through sensors and performs actions through actuators to achieve specific objectives.

In a smart home automation system, intelligent agents control lighting, air conditioning, temperature, door locks, and security alarms.

They use information such as motion detection, temperatures, time of day, and user preferences to make decisions automatically.

For example, when a person enters a room, the system detects their presence and turns on the lights. When everyone leaves, it switches off unnecessary electrical appliances to save energy.

3. Types of intelligent agents

A. Simple reflex agent

A simple reflex agent selects an action based only on the current percept. it uses condition-action rules such as "if condition the action".

it does not maintain a detailed memory of previous events

working principle:

1. Receives input from sensors
2. matches the input with a predefined rule
3. selects the corresponding action
4. performs the action using an actuator.

Examples in a smart home:

1. if motion is detected, turn on the lights
2. if the temperature exceeds a specified limit, turn on the AC.
3. if an unauthorized door opening is detected, activate the alarm.

Advantages:

1. simple to design and implement.
2. Responds quickly to immediate conditions.
3. Requires less memory and computation.
4. Suitable for routine automation tasks

Analysis: A simple reflex agent is useful for basic lighting and alarm functions, but it is insufficient for a complete smart home that must adapt to user behavior.

B. Model-Based agent

A model-based agent maintains an internal state or representation of it's Environment. it uses current sensor information and past observations to Estimate the current situation.

This is particularly useful when the agent cannot observe Everything directly.

working Principle:

1. collects information from sensors.
2. updates its internal model of the Environment.
3. uses the model to understand the current situation.
4. selects an appropriate action.
5. updates the model after receiving new information.